

НАУЧНАЯ СТАТЬЯ | 01 ОКТЯБРЯ 2011 ГОДА

Податливость легких и объем легких у бесхвостых земноводных зависят от экоморфологии: влияние на лимфодренаж при искусственной вентиляции легких **FREE**

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Вертикальное движение лимфы от вентральных отделов к расположенным дорсально лимфатическим сердцам у бесхвостых земноводных осуществляется за счет специализированных скелетных мышц, которые работают в связке с дыхательной системой. Мы предполагаем, что у более наземных видов с большей способностью к мобилизации лимфы и более высокой скоростью лимфотока объем легких и растяжимость легочной ткани будут больше, чем у полуводных или водных видов. Мы измерили *in situ* среднюю и максимальную податливость (Добъём/Давление), растяжимость (%Добъём/Давление) и объём лёгких в диапазоне физиологических давлений (от 1,0 до 4,0 смН₂O) у девяти видов бесхвостых земноводных, представляющих три семейства (Bufonidae, Ranidae и Pipidae), с разным весом тела и средой обитания — от наземной до водной. Мы также изучили взаимосвязь между этими легочными показателями и лимфотоком у полуназемных буфонид (*Rhinella marina*), полуводных безлёгочных лягушек (*Lithobates catesbeianus*) и водных пиповых (*Xenopus laevis*). Аллометрическое масштабирование податливости легких и объема легких в зависимости от массы тела выявило значительные различия на уровне семейств: показатели масштабирования варьировались от 0,75 у жаб до 1,3 у лягушек. В соответствии с нашей гипотезой, у наземных видов семейства жаб податливость легких и объем легких были значительно выше, чем у полуводных видов семейства настоящих лягушек и водных видов семейства квакш. Растяжимость легких варьировалась от 20 до 35 % смН₂O⁻¹ у

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представителей трех семейств, но не коррелировала с экоморфологией. Для трех видов, у которых имеются данные о лимфотоке, у *R. marina* была значительно более высокая

($P < 0,001$) максимальная комплаентность ($84,9 \pm 2,7$ мл $\text{cmH}_2\text{O}^{-1} \text{ кг}^{-1}$) и объем легких ($242,1 \pm 5,5$ мл кг^{-1}) по сравнению с *L. catesbeianus* ($54,5 \pm 0,12$ мл $\text{cmH}_2\text{O}^{-1} \text{ кг}^{-1}$ и $139,3 \pm 0,5$ мл кг^{-1}) и *X. laevis* ($30,8 \pm 0,7$ мл $\text{cmH}_2\text{O}^{-1} \text{ кг}^{-1}$ и $61,3 \pm 2,5$ мл кг^{-1}). Скорость лимфотока также была самой высокой у *R. marina*, самой низкой у *X. laevis* и средней у *L. catesbeianus*. Таким образом, существует тесная взаимосвязь между растяжимостью легких, объемом легких и скоростью лимфотока. Это позволяет предположить, что способность к мобилизации лимфы может частично объяснять различия в растяжимости и объеме легких у бесхвостых земноводных.

Ключевые слова: вентиляция легких, движение лимфы, растяжимость, механика легких

Введение

Легкие бесхвостых земноводных, помимо газообмена, играют важнейшую роль в вокализации, поддержании плавучести в водной среде и защитном поведении ([Хиллман и др., 2009](#)). Недавно мы также описали новую функцию легочной вентиляции — вертикальное перемещение лимфы из вентральных отделов тела в дорсальные лимфатические сердца у бесхвостых земноводных ([Хедрик и др., 2007](#); [Хиллман и др., 2010](#)). Мы предполагаем, что при дефляции легких лимфа движется вертикально за счет эффекта «всасывания» при отрицательном давлении, которое поднимает лимфу от конечностей к субпозвоночному лимфатическому мешку, расположенному над легкими. Эту гипотезу подтверждают несколько фактов. Во-первых, прямое измерение давления в субпозвоночном лимфатическом мешке во время вентиляции показывает, что отрицательное давление, возникающее в субпозвоночном мешке во время выдоха, достаточно для того, чтобы лимфа поступала из вентральных отделов в субпозвоночный мешок, который напрямую связан с задними лимфатическими сердцами ([Хедрик и др., 2007](#)). Во-вторых, при надувании легких сдавливаются лимфатические мешки, окружающие легкие (например, боковые и брюшные мешки) ([Картер, 1979](#)), и лимфа устремляется к лимфатическим сердцам ([Хедрик и др., 2007](#)). В-третьих, введение пластиковой спирали в субпозвоночный мешок механически препятствует полному раздуванию и сдуванию легких и значительно снижает отток лимфы из задних конечностей ([Хиллман и др., 2010](#)). Наконец, вентиляция легких также координируется с сокращением различных скелетных мышц, которые прикрепляются к коже в подмышечных впадинах и уростиле ([Дрюс и др., 2007](#)), что также влияет на отток лимфы ([Хиллман и др., 2010](#)).

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Anurans vary markedly in body shape, from extremely dorso-ventrally flattened aquatic (e.g. *Pipa pipa*) to globose fossorial species (e.g. *Breviceps* spp.) (see [Hillman et al., 2009](#)), and their lungs occupy a large fraction of the pleuroperitoneal space ([Lauridsen et al., 2011](#)). Thus, lung volume relative to body mass should be largest for globose anurans, but nevertheless large for other amphibians, although this has not been systematically examined.

Pulmonary compliance (C) is defined as the change in lung volume per unit change in intrapulmonary pressure ($\Delta V/\Delta P$). Compliance is also the product of distensibility (D ; percent change in volume/change in pressure) and lung volume (V_{lung}), thus differences in body mass and relative lung volume (e.g. globose vs dorso-ventrally flattened) are likely to contribute to species differences in lung compliance. Distensibility is a measure of the surface tension of the air–lung interface and the elastic forces of the lung tissue and body wall, and hence might be similar for different anurans. Although the role of lung compliance and volume to the contribution of lymph movement is unclear, it is possible that the ability of anurans to mobilize lymph is related to pulmonary factors such as pulmonary compliance and lung volume, particularly as lung ventilation is important for lymph movement ([Hedrick et al., 2007](#)).

The lymph flux of anurans is correlated with ecomorphology; that is, terrestrial species have significantly higher lymph flux rates than semi-terrestrial or aquatic species ([Hillman et al., 2011](#)). Because lung ventilation plays a key role in the movement of lymph, we hypothesize that pulmonary compliance and lung volume is also related to anuran ecomorphology, with greater pulmonary compliance and lung volume in more terrestrial species that require greater lymph mobilization. Our hypothesis is that species with higher pulmonary compliance in the physiological range of intrapulmonary pressures should be able to more effectively move lymph and stabilize plasma volume during dehydration. Consequently, compliance should be greater for terrestrial species than for aquatic species.

To our knowledge, the pulmonary compliance of anurans has only been examined previously for two species within the family Ranidae, *Rana* (= *Lithobates*) *pipiens* ([West and Jones, 1975b](#); [Hughes and Vergara, 1978](#)) and *Rana* (= *Lithobates*) *catesbeiana* ([Dupré et al., 1985](#)). These studies compared the inflation/deflation characteristics of anuran lungs with those of mammalian species. In light of our recent studies demonstrating a role for ventilation in the movement of lymph ([Hedrick et al., 2007](#); [Hillman et al., 2010](#)), the present study examined pulmonary compliance and lung volume in selected representatives of three anuran families. The members of these families are found in habitats including terrestrial/semi-terrestrial (Bufonidae), semi-aquatic (Ranidae) and aquatic (Pipidae) ([Hillman et al., 2009](#)). We further examined the relationship between pulmonary variables and lymph flux data that are available ([Hillman et al., 2011](#)) for three species representing these families: *la marina* (semi-terrestrial, xeric), *Lithobates catesbeianus* (semi-aquatic) and *Xenopus laevis*

(aquatic). These species differ in dehydration tolerance (*R. marina*>*L. catesbeianus*=*X. laevis*) ([Hillman, 1978](#); [Hillman, 1987](#); [Hillman et al., 1987](#)) and the degree of development of skeletal muscles associated with lymph flux (*R. marina*>*L. catesbeianus*>*X. laevis*) ([Drewes et al., 2007](#)). Our prediction was that pulmonary compliance and lung volume would show a similar pattern given the importance of lung ventilation to lymph flux in these species.

MATERIALS AND METHODS

Animals

Amphibians were purchased from commercial suppliers. They were maintained at 24°C in conditions appropriate to their natural habitat: water for aquatic species, water and dry areas for the other species. Five species of Bufonidae [green toad, *Anaxyrus debilis* Girard 1854, 3.9–5.2 g, *N*=6; Great Plains toad, *Anaxyrus cognatus* Say 1822, 66.7–109.2 g, *N*=7; red-spotted toad, *Anaxyrus punctatus* Baird & Girard 1852, 14.0–25.2 g, *N*=6; cane toad, *Rhinella marina* (Linnaeus 1758), 39.0–160.0 g, *N*=19; common Asiatic toad, *Duttaphrynus melanostictus* Schneider 1799, 23.7–63.3 g, *N*=10], two species of Ranidae [North American bullfrog, *Lithobates catesbeianus* Shaw 1802, 55.0–151.0 g, *N*=17; southern leopard frog, *Lithobates sphenoccephalus* (Cope 1886), 8.1–48.5 g, *N*=8] and two species of Pipidae [African clawed frog, *Xenopus laevis* Daudin 1802, 25.6–69.2 g, *N*=11; Surinam toad, *Pipa pipa* (Linnaeus 1758), 93.5–204.5 g, *N*=9] were studied. These species were chosen to encompass a large range of body masses to investigate family-level allometric relationships for pulmonary variables. All experiments were conducted at 20°C. All animals were salvaged from experiments that measured apparent lymph volumes in the femoral lymph sac of frogs that were anesthetized in buffered MS-222 (see [Hillman et al., 2011](#)). Following lymph volume measurement, and while anesthetized, the animals were cranially pithed and then decapitated for measurements of pulmonary compliance, distensibility and lung volume as detailed below. All procedures were conducted at Portland State University with IACUC approval (protocol no. 10.01.11.1).

Experimental protocols

For each animal, a flared tracheal catheter, fashioned from polyethylene tubing of appropriate size for the species used, was inserted through the glottis. The catheter was anchored and sealed with a combination of sutures (4-0 silk) and cyanoacrylate adhesive. The catheter was connected to a T-junction, which had a syringe and infusion pump (Harvard Apparatus, model 975, Holliston, MA, USA) on one arm and a volumetric pressure transducer (Grass model PT5A, Braintree, MA, USA) on the other. The pressure transducer was calibrated using static water column pressure. Voltage output from the pressure transducer was digitized and recorded with a data acquisition system (MacLab/4e, ADInstruments, Colorado Springs, CO, USA) and computer. An infusion pump delivered air to the lung at a rate of approximately 20% of body volume min⁻¹. Because inflation was

continuous, the measurements represent dynamic compliance. The compliance of the syringe, tubing and transducer was $0.12 \pm 0.002 \text{ ml cmH}_2\text{O}^{-1}$, and was subtracted from the total pulmonary compliance measurement.

Data analysis and statistics

We chose to determine compliances ($\Delta V/\Delta P$) within approximately 1–4 cmH₂O (0.1–0.4 kPa) because anurans normally experience this range of *in vivo* intrapulmonary pressures over a ventilatory cycle ([DeJongh and Gans, 1969](#); [West and Jones, 1975a](#); [Vitalis and Shelton, 1990](#); [Wang, 1994](#)). The mean compliance over the pressure intervals of 0.5–1.5, 1.5–2.5, 2.5–3.5 and 3.5–4.5 cmH₂O were calculated from each trace, corresponding to means of 1.0, 2.0, 3.0 and 4.0 cmH₂O. Lung inflation was repeated five to 10 times for each individual, and the results are the means of those repeated inflation values (see [Fig. 1](#)). The mean volume change over each of these intervals was derived from the air infusion rate (i.e. 20% of body volume min⁻¹). The mean pulmonary



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liance (

) was the

mean compliance of these four pressure intervals, and the maximal compliance ($C_{\max}$) was the highest compliance measured at the four pressure intervals over which compliances were determined. The volume of air necessary to achieve an intrapulmonary pressure of 4 cm H₂O was also calculated from the pressure trace and reported as total lung volume (V_{lung}). Pulmonary distensibility (D_{pul}) was calculated from the pressure–volume relationship at 1 cmH₂O intervals by dividing the compliance by $C_{\max}$ measured at each 1 cm H₂O interval and is expressed as % cmH₂O⁻¹.

Data are reported as means $\pm$ s.e.m. unless otherwise noted. Standard linear regression analysis and one-way ANOVA were used as appropriate. All data analyses were conducted using GraphPad Prism v. 5 (San Diego, CA, USA).

RESULTS

There were highly significant ($P < 0.001$) correlations ($r^2 = 0.58$ to 0.83) for the allometric relationships



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and V_{lung} for

each family ([Table 1](#); [Fig. 2](#)). Furthermore, there were clear family-level differences in the relationship between pulmonary variables and body mass. Bufonidae had higher intercepts and lower slopes for



$C_{\max}$, and V_{lung} , whereas Pipidae had lower intercepts and higher slopes; Ranidae were intermediate for both intercepts and slopes.

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Table 1. Allometric relationships for maximal pulmonary compliance (C_{max} ; ml cmH₂O⁻¹), mean pulmonary



compliance (; ml cmH₂O⁻¹)
and lung volume (V_{lung} ; ml) for Bufonidae (five species), Ranidae (two species) and Pipidae (two species)

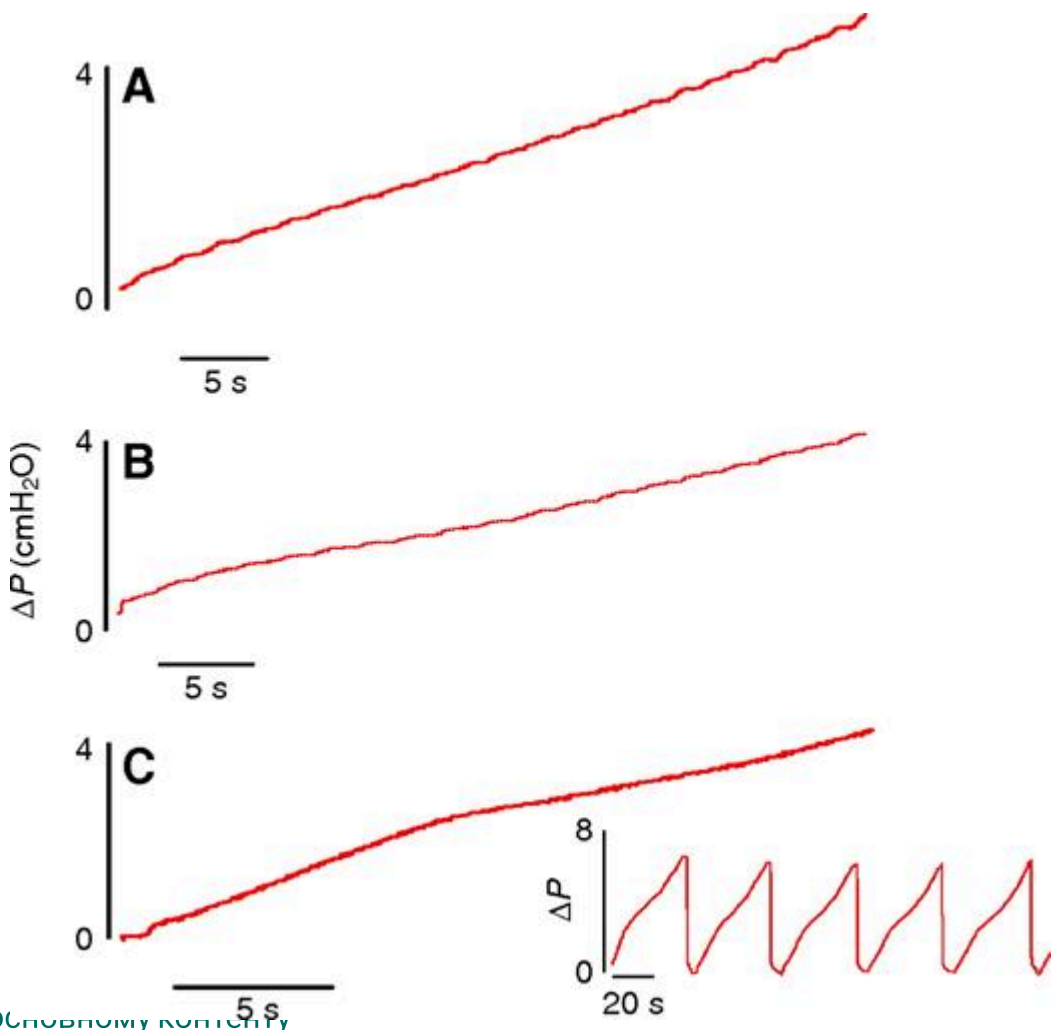


Table 1. Allometric relationships for maximal pulmonary compliance ($C_{\max}$; ml cmH₂O⁻¹), mean pulmonary compliance ($\bar{C}$; ml cmH₂O⁻¹) and lung volume (V_{lung} ; ml) for Bufonidae (five species), Ranidae (two species) and Pipidae (two species)

Family	<i>Y</i>	<i>a</i>	<i>b</i>	<i>P</i>	<i>r</i> ²
Bufonidae	$C_{\max}$	0.263	0.74	<0.0001	0.80
	$\bar{C}$	0.170	0.76	<0.0001	0.83
	V_{lung}	0.724	0.76	<0.0001	0.75
Ranidae	$C_{\max}$	0.064	0.96	<0.0001	0.58
	$\bar{C}$	0.049	0.96	<0.0001	0.73
	V_{lung}	0.182	0.94	<0.0001	0.69
Pipidae	$C_{\max}$	0.015	1.19	<0.0001	0.80
	$\bar{C}$	0.008	1.24	<0.0001	0.81
	V_{lung}	0.018	1.33	<0.0001	0.77

Dependent variables (*Y*) are expressed as $Y=aM^b$, where *a* is the intercept, *M* is body mass (g) and *b* is the mass exponent (see Fig. 2).

Fig. 1.



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Original recordings of pressure changes (ΔP , cmH₂O; 1 cmH₂O=0.1 kPa) over time in the pulmonary system during lung inflation for (A) *Rhinella marina*, (B) *Lithobates catesbeianus* and (C) *Xenopus laevis*. Inset: five separate inflations for one experiment in *X. laevis* illustrate the repeatability of the technique. See Materials and methods for details.

Pulmonary compliance and lung volume measured for the species in this study ([Table 2](#)) represent the mass-specific values based upon family-level allometric equations ([Table 1](#)). Bufonidae species had higher values for $C_{\max}$,



and V_{lung} compared with

Ranidae, which had higher values than Pipidae ([Table 2](#)). Overall, $C_{\max}$ for Bufonidae was 105.1 ± 4.6 ml cmH₂O⁻¹ kg⁻¹ ($N=48$), which was significantly greater ($P<0.001$) than that for Ranidae (55.7 ± 0.3 ml cmH₂O⁻¹ kg⁻¹, $N=28$) and both were significantly greater than that for Pipidae (34.4 ± 1.0 ml cmH₂O⁻¹ kg⁻¹, $N=20$). There was considerable variation in $C_{\max}$ among the Bufonidae species, with values ranging from 83.2 ± 1.6 cmH₂O⁻¹ kg⁻¹ in *A. cognatus* to 177.5 ± 2.2 cmH₂O⁻¹ kg⁻¹ in *A. debilis*. Lung volume followed a similar pattern, with Bufonidae (310.2 ± 13.6 ml kg⁻¹, $N=44$) being significantly

higher ($P<0.001$) than Ranidae ($144.0\pm1.3\text{ ml kg}^{-1}$, $N=20$), both of which were significantly higher ($P<0.001$) than Pipidae ($74.6\pm3.8\text{ ml kg}^{-1}$, $N=20$). Lung volume also exhibited a large variation among the Bufonidae, ranging from $242.1\pm5.4\text{ ml kg}^{-1}$ in *R. marina* to $503.8\pm5.8\text{ ml kg}^{-1}$ in *A. debilis*.

Table 2. Mean mass and pulmonary variables C_{max} ($\text{ml cmH}_2\text{O}^{-1}$),

Family/species	Mass (g)	C_{max} ($\text{ml cm H}_2\text{O}^{-1}\text{ kg}^{-1}$)	$\bar{C}$ ($\text{ml cm H}_2\text{O}^{-1}\text{ kg}^{-1}$)	V_{lung} (ml kg^{-1})
Bufonidae				
<i>Rhinella marina</i>	90.1±10 (19)	84.9±2.7 (19)	58.6±1.9 (16)	242.1±5.4 (15)
<i>Anaxyrus debilis</i>	4.6±0.21 (6)	177.5±2.2 (6)	118.1±1.4 (6)	503.8±5.8 (6)
<i>Anaxyrus cognatus</i>	85.0±6.4 (7)	83.2±1.6 (7)	58.7±1.1 (7)	250.5±4.6 (7)
<i>Anaxyrus punctatus</i>	20.3±2.2 (6)	121.2±3.5 (6)	83.1±2.2 (6)	354.4±9.4 (6)
<i>Duttaphrynus melanostictus</i>	35.1±3.7 (10)	105.5±2.4 (10)	73.1±1.5 (10)	311.6±6.5 (10)
Ranidae				
<i>Lithobates catesbeianus</i>	88.9±5.5 (17)	54.5±0.5 (17)	41.0±0.1 (17)	139.3±0.5 (17)
<i>Lithobates sphenoccephalus</i>	25.0±3.9 (11)	57.6±0.4 (11)	43.3±0.3 (11)	151.3±1.6 (11)
Pipidae				
<i>Xenopus laevis</i>	41.8±5.6 (11)	30.8±0.7 (11)	20.1±0.6 (11)	61.3±2.5 (11)
<i>Pipa pipa</i>	133.8±13.7 (9)	38.7±0.7 (9)	26.8±0.6 (9)	90.9±2.8 (9)

V_{lung} (ml kg^{-1}) measured in five species of Bufonidae, two species of Ranidae and two species of Pipidae

Table 2. Mean mass and pulmonary variables C_{max} ($\text{ml cmH}_2\text{O}^{-1}\text{ kg}^{-1}$), $\bar{C}$ ($\text{ml cmH}_2\text{O}^{-1}\text{ kg}^{-1}$) and V_{lung} (ml kg^{-1}) measured in five species of Bufonidae, two species of Ranidae and two species of Pipidae

means ± s.e.m. (no. of animals).

There was no significant allometric relationship for D_{pul} and body mass for Bufonidae, Ranidae or Pipidae at any lung pressure, except in two cases: D_{pul} for Bufonidae at pulmonary pressures of 3 and 4 cmH₂O were significantly related to body mass ($P < 0.02$ and < 0.01 , respectively). However, body mass explained very little of the variation in D_{pul} ($r^2 = 0.11$ at 3 cmH₂O and $r^2 = 0.15$ at 4 cmH₂O) (data not shown). Because there was no significant allometric relationship with D_{pul} , we examined family-level differences in D_{pul} at each pulmonary pressure. We did find significant differences in D_{pul} at 2, 3 and 4 cmH₂O, but these differences were not consistent with differences in ecomorphology. For example, at 2 cmH₂O, D_{pul} for Ranidae was $32.6 \pm 1.0\%$ cmH₂O⁻¹ and was significantly higher ($P < 0.005$) than that for either Bufonidae ($29.2 \pm 1.1\%$ cmH₂O⁻¹) or Pipidae ($26.1 \pm 1.4\%$ cmH₂O⁻¹) (Fig. 3). D_{pul} for Pipidae was significantly higher ($P < 0.01$) than that for either Bufonidae or Ranidae at 3 and 4 cmH₂O (Fig. 3). However, the differences in D_{pul} at these lung pressures were small and were revealed because of the large sample size and low variance of D_{pul} (Fig. 3).

The relationships between C_{max} ,



and V_{lung} measured in

this study and lymph flux measured for *R. marina*, *L. catesbeianus* and *X. laevis* ([Hillman et al., 2011](#)) were significant for the femoral ([Fig. 4](#)) and subvertebral ([Fig. 5](#)) sac lymph for all three species. The consistent pattern was that lymph flux and each pulmonary variable was significantly higher ($P<0.001$) for the most terrestrial species (*R. marina*), the lowest for the aquatic species (*X. laevis*) and intermediate for the semi-aquatic species (*L. catesbeianus*).

DISCUSSION

Consistent with our hypothesis, pulmonary compliance and lung volume were correlated with habitat type (ecomorphology) for the nine species of anurans examined in this study. The terrestrial and semi-terrestrial Bufonidae species had the highest values for $C_{\max}$,



and V_{lung} , the semi-

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aquatic Ranidae species had intermediate values, and the aquatic Pipidae had the lowest values.

The pulmonary values were also strongly correlated with the ability to mobilize lymph for three representative species, with higher lymph fluxes in *R. marina*, followed by *L. catesbeianus* and *X. laevis*

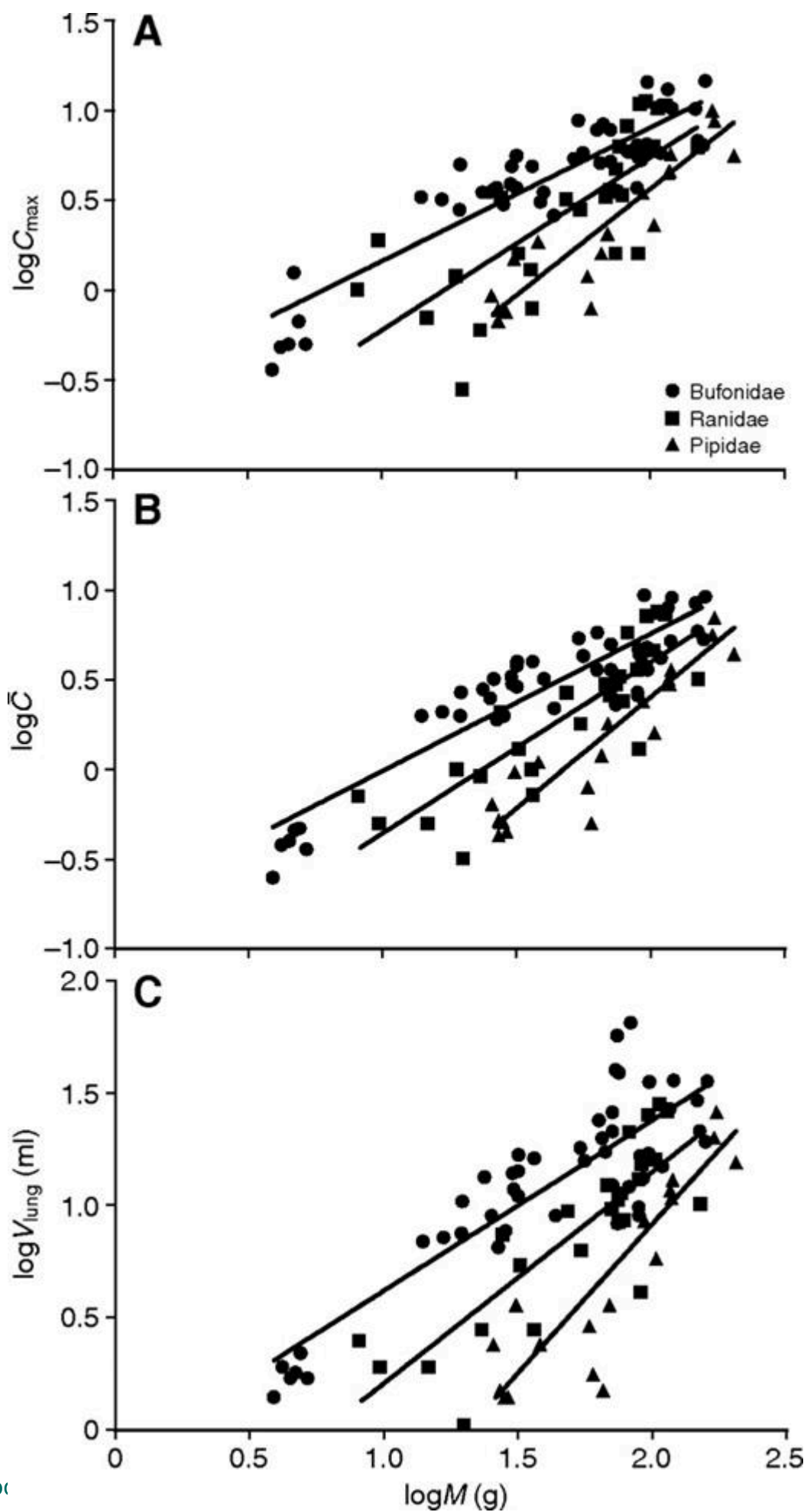
([Hillman et al., 2011](#)) (Figs [4](#), [5](#)). These patterns suggest that variation in ecomorphology and pulmonary factors such as compliance and volume may, in part, account for some of the variation in lymph mobilization capacity.

Pulmonary compliance, distensibility and lung volume

Previous work has examined pulmonary compliance for only two Ranidae species (*Lithobates pipiens* and *L. catesbeianus*). Maximum mass-specific pulmonary compliance of adult *L. pipiens* during lung inflation ranged from 22 to 49 ml cmH₂O⁻¹ kg⁻¹ ([West and Jones, 1975b](#); [Hughes and Vergara, 1978](#)), which is slightly lower than our values for *L. catesbeianus* and *L. sphenoccephalus* ([Table 2](#)). Because we did not measure pulmonary compliance for *L. pipiens*, these differences may be due to variation between species. The values from previous studies are total pulmonary compliance (i.e. lungs and body wall measured *in situ*) and are therefore directly comparable to the values obtained in this study. Dupré et al. measured volume-specific compliance of isolated lungs during development for bullfrogs, but did not provide values for body mass ([Dupré et al., 1985](#)). Their value for lung compliance, measured in air, was approximately 35 ml cmH₂O⁻¹ for adult *L. catesbeianus*. Our measurement of absolute $C_{\max}$ *in situ* was 6.2 ± 0.8 ml cmH₂O⁻¹ for *L. catesbeianus*, approximately five times lower than the value for isolated lungs. Because our measurement included the lungs and body wall, we would expect our values for $C_{\max}$ to be significantly less than the compliance of isolated lungs. Body wall compliance (C_B) can be estimated as $C_B^{-1} = C_T^{-1} - C_L^{-1}$, where C_T is total pulmonary compliance ($=C_{\max}$) and C_L is compliance of the isolated lungs. Combining the data for C_L from Dupré et al. ([Dupré et al., 1985](#)), we calculate a C_B of 7.5 ml cmH₂O⁻¹ for *L. catesbeianus*, indicating that compliance of the body wall is nearly equal to the total respiratory system compliance. A previous study of the Tokay gecko (*Gekko gecko*) also found that C_T and C_B were approximately equal, owing to the high C_L ([Milsom and Vitalis, 1984](#)).

Fig. 2.





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Allometric relationships for (A) maximal pulmonary compliance, (B) mean pulmonary compliance and (C) lung volume for species of Bufonidae (*Rhinella marina*, *Anaxyrus debilis*, *A. cognatus*, *A. punctatus* and *Duttaphrynus melanostictus*), Ranidae (*Lithobates catesbeianus* and *L. sphenoccephalus*) and Pipidae (*Xenopus laevis* and *Pipa pipa*) used in this study. Equations describing the relationships between $C_{\max}$,



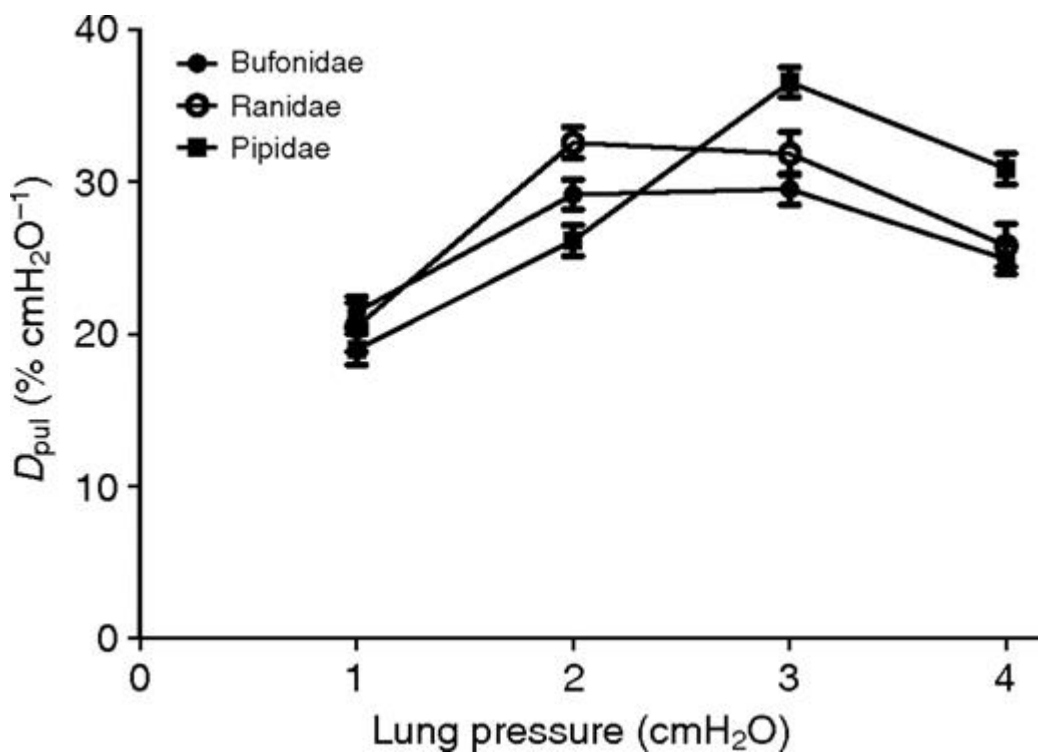
and V_{lung} as

a function of body mass are given in [Table 1](#). M , body mass.

Fig. 3.

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Pulmonary distensibility (D_{pul} , % cmH₂O⁻¹) as a function of lung pressure (cmH₂O) for the representative species of Bufonidae, Ranidae and Pipidae used in this study. For each group, D_{pul} is maximal at intermediate lung pressures with lower distensibility at low and high lung pressures.

Values are means $\pm$ s.e.m.

Pulmonary compliance of the anurans in this study and from previous studies ([Hughes and Vergara, 1978](#); [Dupré et al., 1985](#)) is approximately an order of magnitude greater than for mammals of comparable body mass ([Stahl, 1967](#)). Indeed, our values for Bufonidae species, and for *A. debilis* in particular, are among the highest mass-specific pulmonary compliance and lung volume values reported for vertebrates. Mass-specific pulmonary compliance and lung volume is also much higher for reptiles than mammals ([Tenney and Tenney, 1970](#); [Perry and Duncker, 1978](#); [Perry and Duncker, 1980](#); [Milsom and Vitalis, 1984](#); [Bartlett et al., 1986](#); [Lutcavage et al., 1989](#); [Klein et al., 2003](#)). Reptiles also have a very large variation in total pulmonary system compliance ranging from approximately 8 ml cmH₂O⁻¹ kg⁻¹ for the turtle *Pseudemys scripta* ([Vitalis and Milsom, 1986](#)) to 466 ml cmH₂O⁻¹ kg⁻¹ for the chameleon *Chamaeleo chamaeleon* ([Perry and Duncker, 1978](#)). Whether this large variation in pulmonary compliance represents family-level differences, as we have shown with anurans, is not clear. It has been suggested that the high pulmonary compliance and lung volume of reptiles also represents a functional compromise between gas exchange and other important functions such as buoyancy and display ([Perry and Duncker, 1978](#)).

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Although we have shown that pulmonary compliance and lung volume are well correlated with the

degree of terrestriality for anurans, gas exchange does not show the same correlation for these

species. Maximal oxygen consumption ($V_{O_{2,max}}$) from 16 studies of Bufonidae was 0.97 ± 0.06 ml O₂ g⁻¹

h^{-1} , compared with $0.50 \pm 0.05 \text{ ml O}_2 \text{ g}^{-1} \text{ h}^{-1}$ for Ranidae (17 studies) and $0.77 \pm 0.08 \text{ ml O}_2 \text{ g}^{-1} \text{ h}^{-1}$ for Pipidae (four studies) ([Gatten et al., 1992](#)). If gas exchange were the principal driving force for the variation in lung volume, we would expect species with the highest metabolic rates to also have the highest values for lung volume. Because this was not the case, the results of the present study suggest that habitat preference, degree of terrestriality and lymph mobilization capacities may explain more of the variation in pulmonary compliance and lung volume. However, it will be necessary to investigate these relationships for species representing a broader array of families, using a more detailed phylogenetic analysis, before any firm conclusions can be drawn.

Unlike pulmonary compliance and lung volume, pulmonary distensibility was not consistently correlated with body mass. We found that D_{pul} varied slightly as a function of lung pressure, but the variation was not consistent with the ecomorphology of these species. The lungs of anurans comprise relatively simple sac-like structures in some species or have 'sac alveoli' bounded by primary, secondary or tertiary septa that increase the respiratory surface area ([Tenney and Tenney, 1970](#); [Goniakowska-Witalinska, 1986](#); [Maina, 1989](#); [Maina, 2002](#)). Pipidae have lower lung septation compared with Ranidae and Bufonidae ([Tenney and Tenney, 1970](#)), and this may account for our finding that D_{pul} was higher for Pipidae at higher pulmonary pressures. The ratio of total lung septal area to total lung surface area is roughly equal for Bufonidae and Pipidae, but lower in Ranidae ([Czopek, 1965](#)). It is likely that such variations in lung structure account for the variation in D_{pul} between the three families in the present study. A previous study found that volume-specific lung compliance did not vary significantly during development for *L. catesbeianus* ([Dupré et al., 1985](#)). Because pulmonary compliance is the product of distensibility and lung volume, and D_{pul} was essentially constant, our data indicate that variation in pulmonary compliance is primarily a function of differences in lung volume.

It has been noted previously that V_{lung} and pulmonary compliance are larger for amphibians compared with other vertebrates ([Tenney and Tenney, 1970](#); [Hughes and Vergara, 1978](#)). Maximal lung volume scaled to body mass with a slope of 1.05 for a number of different amphibians ([Tenney and Tenney, 1970](#)). The slopes for V_{lung} in our study ranged from 0.76 for Bufonidae to 1.33 for Pipidae, which encompasses the range found previously. It should be noted that the allometric slopes obtained previously were for dried, isolated lungs inflated to a pressure of 20 cmH₂O, well outside the normal physiological range of lung pressures ([Tenney and Tenney, 1970](#)). However, despite the very different methods for measuring volume, V_{lung} scaled similarly as a function of body mass in our study and previous studies.

Lung volume measured in previous studies also exhibits considerable variation. For example, V_{lung} of *L. catesbeianus* ranges from approximately 80 to 100 ml kg⁻¹ for awake animals ([Tenney and Tenney, 1970](#); [Glass et al., 1981](#)), but can reach as high as 300 ml kg⁻¹ during anoxia ([Tenney and Tenney, 1970](#)).

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1970). Maximal V_{lung} of *L. pipiens* ranged from 35 ml kg⁻¹ for awake animals ([West and Jones, 1975b](#)) to approximately 210 ml kg⁻¹ for anesthetized animals with artificial lung inflation ([Hughes and Vergara, 1978](#)). Our values for two species of Ranidae (139–151 ml kg⁻¹) encompass previous values for anesthetized and awake animals, indicating that our measurements from pithed animals were not substantially different.

The role of pulmonary compliance and lung volume in lymph movement

We have shown previously that lung ventilation is important in anurans for lymph movement ([Hedrick et al., 2007](#); [Hillman et al., 2010](#)). In the present study, we found that the more terrestrial species (Bufonidae) have relatively larger lungs (cf. [Lauridsen et al., 2011](#)) than either Ranidae or Pipidae, and are likely to be more effective at transmitting pressure to the surrounding lymph sacs during lung inflation. Inflation of the lungs in anesthetized animals causes increases in pressure of the lymph sacs surrounding the lungs ([Hedrick et al., 2007](#)), thus lung inflation would facilitate movement of lymph toward the lymph hearts by 'squeezing' lymph sacs against the body wall.

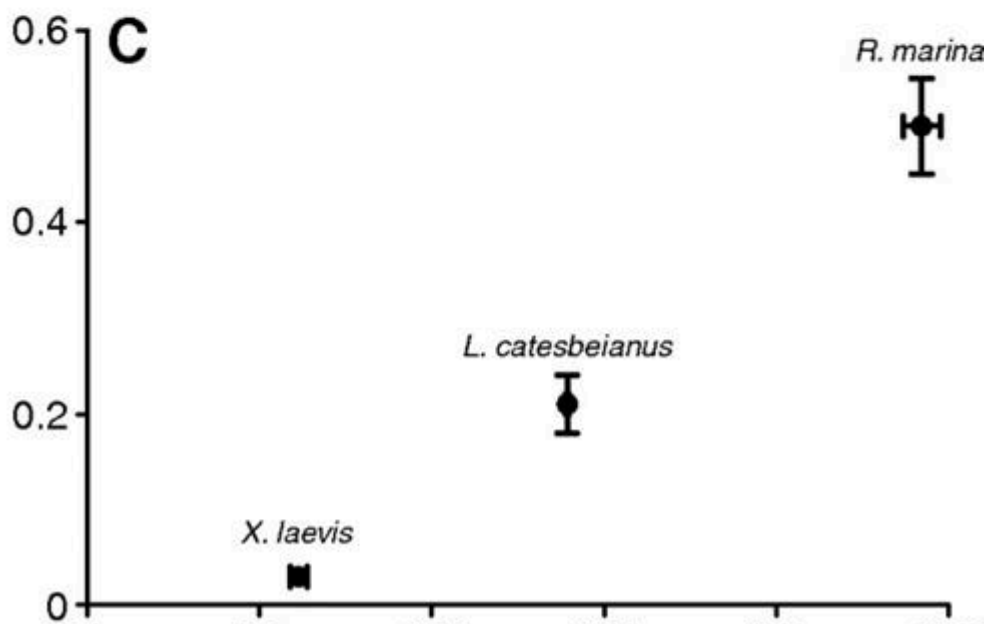
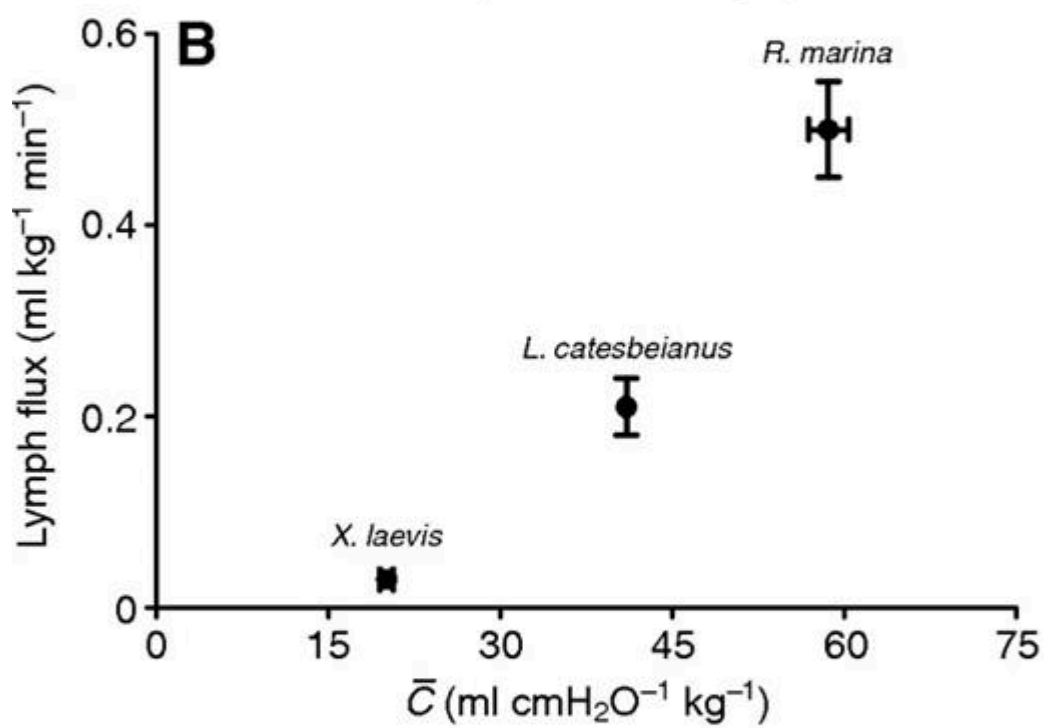
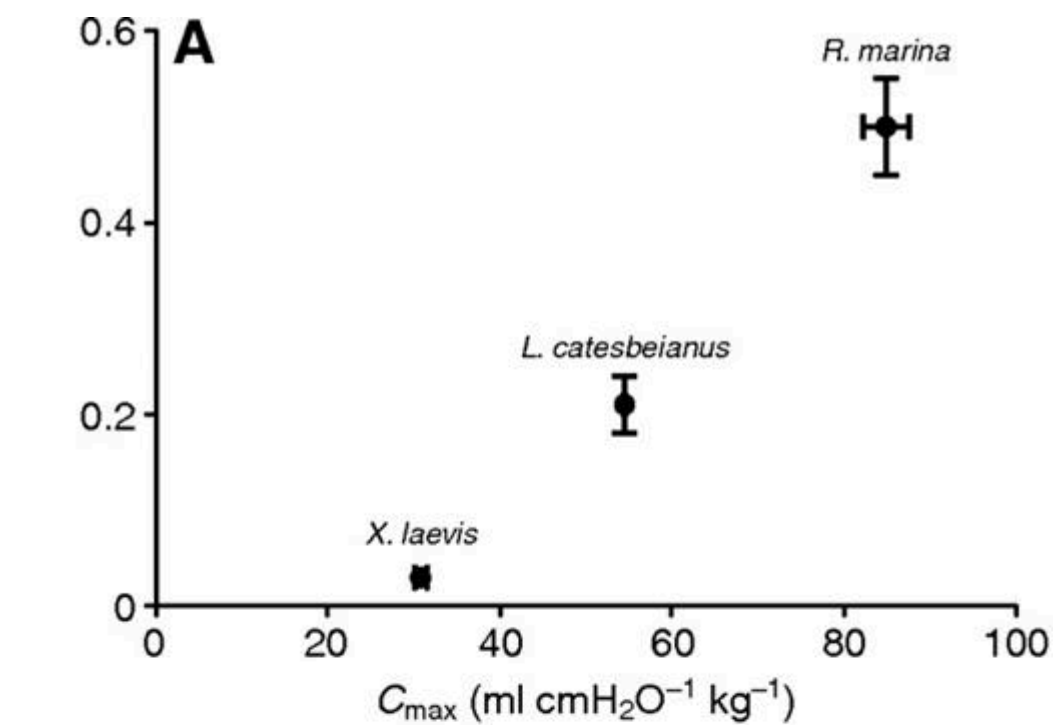
We hypothesize that during lung deflation when lung volume decreases, dorsally located lymph sacs, such as the subvertebral and pubic sacs ([Kampmeier, 1969](#); [Carter, 1979](#)), expand, thus creating a negative lymphatic pressure that serves to 'suction' lymph from the forelimb and hindlimb lymph sacs. The forelimb and hindlimb lymph sacs are connected to subvertebral and pubic sacs via one-way valves [see fig. 3 in Hillman et al. ([Hillman et al., 2011](#))]. In support of this hypothesis, negative pressures measured in the subvertebral and pubic lymph sac are of sufficient magnitude to move lymph vertically during lung deflation in *R. marina* and *L. catesbeianus* ([Hedrick et al., 2007](#)). In addition, interfering with the lungs' ability to inflate and deflate normally by placing a plastic 'coil' in the subvertebral sac of awake *R. marina* significantly reduced lymph flux from the hindlimb femoral sac ([Hillman et al., 2010](#)). The reduction in lung volume and, therefore, pulmonary compliance, clearly interferes with the mechanism involved in the vertical movement of lymph from ventral to dorsal locations. The more terrestrial species, *R. marina*, generates greater negative pressures in the pubic lymph sac than does *L. catesbeianus* ([Drewes et al., 2007](#)), and this is associated with higher lymph flux rates in *R. marina* ([Hillman et al., 2011](#)) (Figs 4, 5). Because changes in lung volume result in negative pressures in lymph sacs that assist in lymph movement, a more compliant lung should produce a greater change in volume for a given change in pressure during exhalation. For example, if we compare pulmonary compliance for 100 g animals from each family using the equations in [Table 1](#), C_{max} is 7.85 ml cmH₂O⁻¹ for Bufonidae, 5.0 ml cmH₂O⁻¹ for Ranidae and 3.6 ml cmH₂O⁻¹ for Pipidae. It is clear that with a larger compliance and lung volume, an equivalent reduction in lung pressure will result in a larger change in lung volume for Bufonidae than for either Ranidae or Pipidae. The larger change in lung volume would translate into a larger change in lymph sac volume and pressure. Thus,

we hypothesize that more compliant lungs, associated with more terrestrial species, are more effective for lymph mobilization than smaller, less compliant lungs.

Fig. 4.

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0 50 100 150 200 250

V_{lung} (ml kg⁻¹)

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Relationship between femoral lymph sac flux and mass-specific pulmonary variables for three species of anurans. (A) Maximal pulmonary compliance (C_{max}); (B) mean pulmonary compliance (



); (C) lung

volume (V_{lung}). Significant differences in all variables between species with *R. marina* > *L. catesbeianus* > *X. laevis*. Values are means $\pm$ s.e.m. Lymph flux values are taken from Hillman et al. ([Hillman et al., 2011](#)).

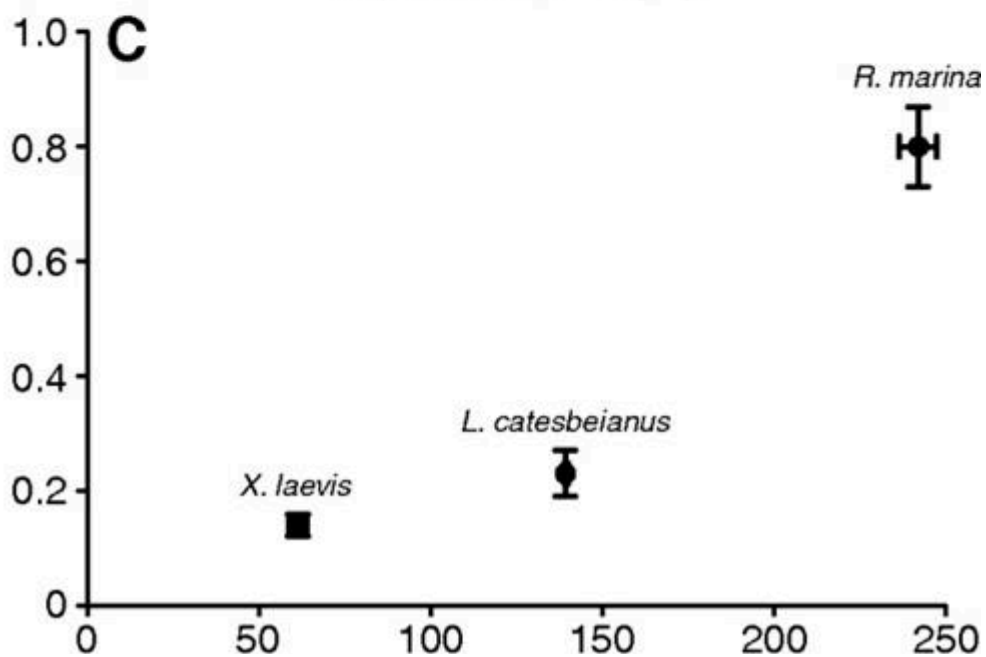
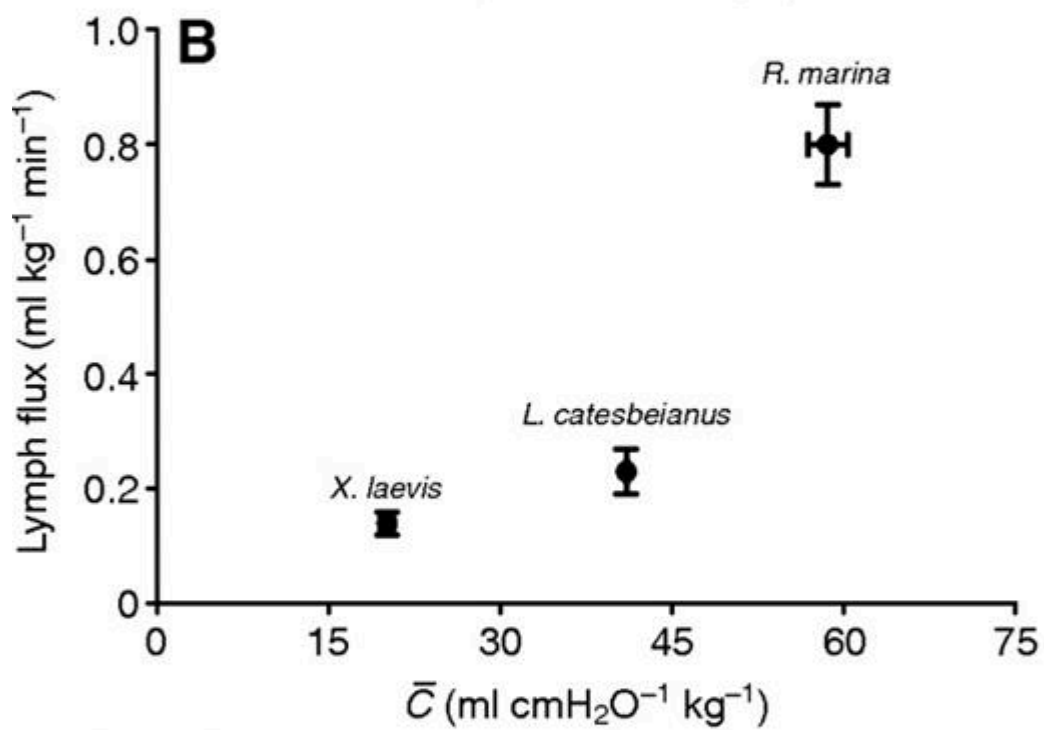
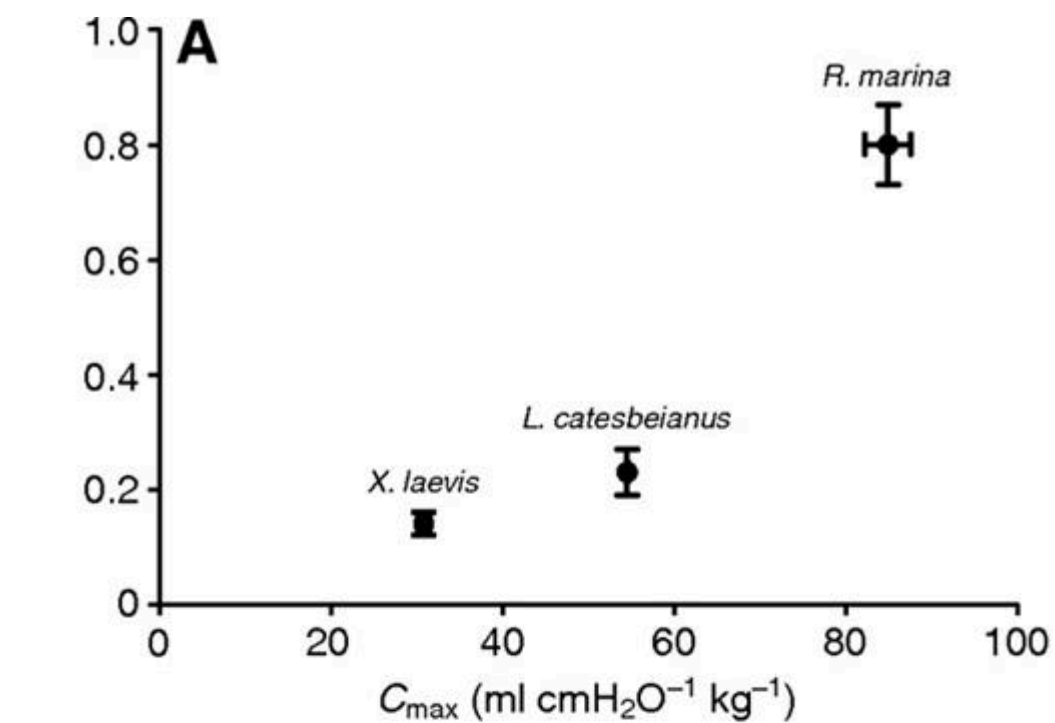
Another key component of this model is likely the large difference between C_L and C_B (see above), which facilitates volume expansion of the subvertebral lymph sac. The subvertebral lymph sac is [Перейти к основному контенту](#) bounded dorsally by the rigid backbone and is ventrally attached to the lung surface by a thin  brane. During expiration, when lung volume decreases and pressure decreases, the relatively  body wall (lower compliance) compared with the lungs and lymph sacs would allow the

expansion of the subvertebral sac overlying the lungs. The expansion of the subvertebral sac creates the negative pressure necessary to move lymph from the limbs to the subvertebral space. However, more work is necessary to explicitly test the role of compliance and volume in the movement of lymph in anurans. Overall, our results suggest that pulmonary compliance and lung volume may play important roles in ventilatory-assisted lymph flux.

Fig. 5.

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V_{lung} (ml kg⁻¹)

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Relationship between subvertebral lymph sac flux and pulmonary variables for three species of anurans. (A) Maximal pulmonary compliance (C_{max}); (B) mean pulmonary compliance (



); (C) lung

volume (V_{lung}). Values are means $\pm$ s.e.m. Lymph flux values are taken from Hillman et al. ([Hillman et al., 2011](#)).

FOOTNOTES

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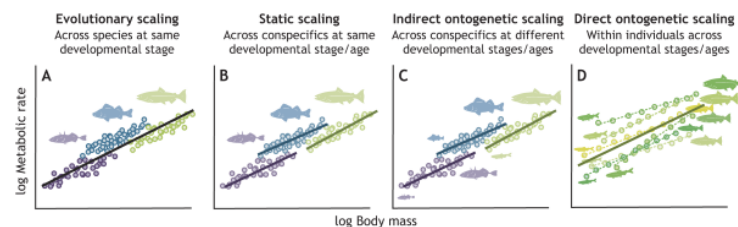
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